MILITARY SPECIFICATION

ARMING WIRE ASSEMBLIES

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers arming wire assemblies for use as arming elements for aircraft dropped mines.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-20	Bag, Cotton, Mailing
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiberboard

Military

MIL-P-116	Preservation-Packaging, Methods of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1325

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STANDARDS

Military

MIL-STD-105	Sampling Tables and Procedures for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
DOD-STD-2101	Classification of Characteristics

DRAWINGS (Naval Sea Systems Command)

PL 3026313	Arming Wire Mark 4 Mod 2
DL 3014527	Shipping Condition, Arming Wire, Mark 4 Type

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117-73	Standard Method of Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3).

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3.3 Materials. Materials shall be in accordance with the applicable drawings (see 2.1).

3.4 Fabrication. Each arming wire assembly shall be in accordance with PL 3026313.

3.5 Performance characteristics.

3.5.1 Pull. When tested as specified in 4.5.3, the swivel assembly shall be operable after submission to a load of 100 pounds, minimum, for five (5) seconds.

3.5.2 Dynamic load (M101). When tested as specified in 4.5.4, the swivel assembly shall not separate or break when submitted to the dynamic load of a five (5) pound weight dropped twenty (20 inches).

Note: Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as major or minor in accordance with DOD-STD-2101. Tests and examinations that verify major characteristics are identified by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

3.5.3 Corrosion. When tested as specified in 4.5.5, the swivel assembly shall be operable after submission to salt spray testing for forty (40) hours, minimum.

3.5.4 Spring force. When tested as specified in 4.5.6, the safety clip shall not release a wire 0.064 inch in diameter when a force of two (2) pounds is applied to the clip in the release direction.

3.5.5 Stripping force. When tested as specified in 4.5.7, the force required to strip a safety clip from a wire 0.064 inch in diameter shall be not less than four (4) pounds or more than fifteen (15) pounds.

3.6 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner. They shall be free of dirt, grease, corrosion, and foreign matter and of damage such as kinks, bends, burrs, or any deformation which would impair its intended use (see 4.5.1 and 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements

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specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Quality control system. The contractor shall provide and maintain an effective quality control system acceptable to the Government covering the supplies and services being procured.

4.1.2 Government verification. All quality assurance operations performed by the supplier will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system are being properly applied, (b) Government source inspection to measure quality of product to be offered for acceptance, and (c) Government inspection of samples for dynamic test to assure compliance with this specification and as specified in the contract or order.

4.2 Classification of inspection. Inspections shall be classified as qualification inspections or quality conformance inspections as defined below:

a. First article inspection. (See 4.3)

b. Quality conformance inspection. (See 4.4)

4.3 First article inspection.

4.3.1 First article. Unless otherwise specified in the contract (see 6.2), a first article of 20 finished arming wire assemblies and 20 complete sets of parts shall be manufactured and prepared for delivery using the methods and procedures proposed for the production lot. The first article shall be prepared for delivery in accordance with Section 5 herein and will be tested for approval as specified in Section 4 herein. The first article is for the purpose of determining that prior to starting production the contractor's production methods are capable of producing assemblies that comply with this specification and the design. The requirements for a first article may be waived by the contracting officer if satisfactory production of the assemblies has been recently accomplished by the contractor.

4.3.2 First article testing. The first article assemblies and parts shall be subjected to the examinations and tests specified in Table I at any activity designated by the procuring agency. Further production by the contractor, prior to approval of the first article, shall be at the contractor's risk.

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TABLE I. First article inspections.

Examinations and Tests	Paragraph	Samples
Visual examination	4.5.1, 4.5.2	20
*Pull test of swivel assembly	4.5.3	20
*Dynamic load test of swivel assembly	4.5.4	20
*Corrosion test of swivel assembly	4.5.5	20
*Spring force of safety clip	4.5.6	20
*Stripping force of safety clip	4.5.7	
Phase I		10
Phase II		10

*Indicates destructive tests not required to be performed by the contractor. These tests are to be performed by the designated Government test activity (see 6.2).

4.3.3 First article approval. First article approval will be granted when the Government testing activity, through testing and subsequent analysis of the test results, is assured of contractor conformance to the contract technical requirements. A first article may be disapproved when either nonconforming characteristics or unauthorized materials are found. Conditional acceptance may be granted at the option of the Government based on analysis of detected defects.

4.4 Quality conformance inspection.

4.4.1 Production lot sample testing.

4.4.1.1 Sampling plan. Unless otherwise specified, the sampling plans and procedures used in determining the acceptability of products shall be in accordance with MIL-STD-105. Each lot submitted for acceptance shall be evaluated for the requirements of Section 3.

4.4.1.1.1 Classification of defects. Defects shall be classified as specified in Table II.

4.4.1.2 Production lot sample testing. A production lot sample shall be selected randomly by the Government inspector from each production lot in accordance with MIL-STD-105, Single Sample Normal Inspection, Inspection Level II. The sample shall be forwarded to the Government-designated testing activity(ies) specified in the contract (see 6.2). Testing shall consist of the examinations and tests of 4.5. At the option of the Government all or a part of the sample may be subjected to the tests of Table III.

4.4.1.3 Acceptance/Rejection criteria. A lot will be rejected when the number of defective samples exceeds the maximum Acceptable Quality Level (AQL) allowed in MIL-STD-105 Single Sample Normal Inspection, Inspection Level II, AQL 1.0 for majors and 2.5 for minors. The lot will require screening and repair or replacement of any defective units.

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TABLE II. Classification of defects (examination) in accordance with MIL-STD-105.

Defect	Major	Minor
Materials failing to meet specified requirements. Dimensions not within specified tolerances. Finish failing to meet specified requirements. Marking missing, incorrect, insufficient, illegible, or not permanent. Workmanship not as specified. Packaging failing to meet specified requirements.	Defects which might impair performance, reliability, maintainability, or safety.	Defects which would not materially reduce performance, reliability, maintainability, or safety.

TABLE III. Production tests.

Examination and Tests	Paragraph	Samples
Visual examination	4.5.1, 4.5.2	In accordance with 4.4.1.1 and 4.4.1.2
*Pull test of swivel assembly	4.5.3	
*Dynamic load test of swivel assembly	4.5.4	
*Corrosion test of swivel assembly	4.5.5	
*Spring force of safety clip	4.5.6	
*Stripping force of safety clip	4.5.7	

*Indicates destructive tests not required to be performed by the contractor. These tests are to be performed by the designated Government test activity (see 6.2).

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4.5 Examinations and tests.

4.5.1 Visual examination. Each arming wire assembly and part shall be examined for conformance with PL 3026313 and this specification and to verify that there are no burrs, sharp edges, kinks, bends, deformations or contaminants (see 3.4.2 and 3.6).

4.5.2 Examination of preservation-packaging. The preservation-packaging shall be examined for conformance with Section 5 and the contract (see 6.2).

4.5.3 Pull test of swivel assembly. The fixture of this test shall have a nineteen thirty-seconds (19/32) inch diameter hardened steel pin for support of the assembly through the large loop. A hook or pin shall be inserted in the small loop and, using the hook or pin, the required load shall be applied and held for the specified period of time. After test, the small loop shall be held in one hand and the large loop shall be struck sharply with a finger of the other hand. Failure of the large loop to spin one half (1/2) turn or more shall class that unit defective (see 3.5.1).

4.5.4 Dynamic load test of swivel assembly. The fixture for this test shall have a pin for support of the large loop of the same configuration as the pin of 4.5.3. The balance of the fixture shall consist of a bar approximately thirty (30) inches long which is slotted at one end to accept the small loop of the swivel. The slotted end shall be drilled through to accept a pin which will attach the swivel to the bar. The other end of the bar shall have a steel disc attached firmly thereto. A five (5) pound weight, bored to slide freely on the bar shall supply the specified load. For testing, the large loop shall be assembled to the firmly supported pin. The slotted end of the bar shall be assembled to the small loop. The weight shall be raised until its bottom surface is twenty (20) inches from the top of the disc. The weight shall then be released to drop freely. Any unit which separates or breaks during test shall be classed defective (see 3.5.2).

4.5.5 Corrosion test of swivel assembly. The loop and swivel assembly shall be tested in accordance with ASTM B117-73. Any assembly failing the spin requirement of 4.5.3, after test, shall be classed defective (see 3.5.3).

4.5.6 Spring force of safety clip. A new .064 inch diameter arming wire conforming with this specification, shall be inserted into, and be held by, the clip. While holding the clip fixed, the specified force shall be applied to the clip in the releasing direction with an appropriate spring scale. Any clip releasing the wire upon application of that force shall be classed defective (see 3.6.1).

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4.5.7 Stripping force of safety clip. The fixture for this test shall consist of a horizontal support with vertical holes or a slot capable of holding the desired number of test samples. The loads shall be applied with an accurate spring scale. The test shall be conducted in two (2) phases. Phase I samples shall be tested immediately after assembly of the clips to the wires. Phase II samples shall be tested at least forty-eight (48) but not more than sixty-four (64) hours after assembly. New .064 inch diameter wires conforming to this specification shall be held in the support and the clip assembled three (3) inches, minimum, from the end. The spring scale shall be attached to the loop in the end of the wire. Any clip stripping from the wire due to a load of four (4) pounds or less shall be classed defective. If stripping does not occur the load shall be gradually increased to fifteen (15) pounds. If stripping has not occurred when the fifteen (15) pound load is attained, the clip shall be classed defective.

5. PACKAGING

5.1 Unit pack.

5.1.1 Arming wire. Each arming wire shall be preserved Method III of MIL-P-116, and attach the appropriate identification tag, Naval Sea Systems Command PL 3026313.

5.1.2 Safety clip. When shipped with arming wire, three hundred safety clips shall be preserved Method III of MIL-P-116 utilizing cotton bag conforming to PPP-B-20.

5.1.2.1 When shipped separately, fifty (50) safety clips shall be preserved Method III of MIL-P-116, utilizing containers conforming to PPP-B-636, weather resistant.

5.2 Intermediate pack.

5.2.1 Arming wire assemblies. Arming wire assemblies, of like items, as unit packed in accordance with 5.1.1 together with three hundred safety clips as prepared in accordance with 5.1.2 shall be packaged in accordance with Naval Sea Systems Command DL 3014527.

5.3 Packing. (See 6.2).

5.3.1 Level A.

5.3.1.1 Arming wire assemblies. Arming wire assemblies, of like items, as intermediate packed in accordance with 5.2.1 shall be packed in wooden boxes conforming to PPP-B-621, Class 2, Style 2 or 4 or plywood boxes conforming to PPP-B-601, overseas type.

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5.3.1.2 Safety clips. Safety clips as unit packed in accordance with paragraph 5.1.2.2 shall be overpacked in wooden boxes conforming to PPP-B-621 Class 2 or plywood boxes conforming to PPP-B-601, overseas type.

5.3.2 Level B.

5.3.2.1 Arming wire assemblies. Arming wire assemblies, of like items, as intermediate packed in accordance with 5.2.1 shall be packed in fiberboard boxes conforming to PPP-B-621 Class 1 or plywood boxes conforming to PPP-B-601, domestic type.

5.3.2.2 Safety clips. Safety clips as unit packed in accordance with 5.1.2.2 shall be overpacked in fiberboard boxes conforming to PPP-B-636.

5.3.3 Commercial.

5.3.3.1 Arming wire assemblies. Arming wire assemblies of like items, as intermediate packed in accordance with 5.2.1 shall be packed in accordance with commercial practice.

5.3.3.2 Safety clips. Safety clips packed in accordance with paragraph 5.1.2.2 shall be packed in accordance with commercial practice.

5.4 Marking.

5.4.1 In addition to the special markings required by the contract or order, unit, intermediate pack, and shipping container shall be marked in accordance with MIL-STD-129 and as required by the drawings listed on DL-3014527.

6. NOTES

6.1 Intended use. The arming wire assemblies and parts covered by this specification are intended to render weapons safe during handling and transit and to subsequently arm the weapons during deployment. Reliability and smooth, consistent operation are therefore imperative.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

a. Title, number and date of this specification.

b. Levels of preservation, packaging and packing.

c. For service or non-service use.

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d. Designated Government test activity.

e. Shipment of safety clips (see 5.1.2).

6.3 MIL-A-003812B(MU) was cancelled 11 March 1980. This specification replaces that specification to satisfy a requirement of the Department of the Navy.

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